

# Reveal the Secrets of Olympics Medals through Data Analysis

## Summary

The Summer Olympics, a prestigious global event since 1896, sees constantly changing medal counts. This paper develops models to **forecast the 2028 medal counts** and explores interesting trends and factors affecting these counts. We started by **cleaning and reorganizing** the provided data, addressing errors and missing information to prepare for model building.

To predict future medal counts, we built a **3D GSRF-based multiple linear regression model**. The model first calculates the medal counts for each sport and then sums them. The **x-value** represents the country's relative competitiveness, derived from the **GSRF model** using past performances. The **y-value** is determined through **TOPSIS** using short-term, up-to-date data on athletes, and the **z-value** is the **actual medal counts**. The model can predict both **gold** and **total medals** by adjusting the z-value and the weights from **TOPSIS** and **AHP**. The model shows a strong goodness of fit ( $R^2 = 0.9067957932583723$ ). Using this model, we predicted the **2028 Los Angeles Olympic medal table** and estimated **prediction intervals**. A **rating model** combining **linear regression** was also used to compare the **2024 and 2028 performances** of countries. We predicted **GBR** to **improve**, while **UZB** may perform **worse**. Additionally, we analyzed each sport in detail. We used **linear regression** and **Shapley values** to explore the relationships between countries' medal counts and specific sports. After **removing outliers** with the **z-score method**, we confirmed a positive linear relationship between total medals and the number of sports. **Important sports** for various countries were identified using **AHP** and **Shapley values**. We also quantified the impact of **home country event choices** through **linear regression** and the **rating model**, incorporating this into the 2028 predictions.

Next, we developed two models to explore **change points** in medal counts linked to the "great coach" effect. Using the **CUSUM method**, we detected **change points** and studied the correlation between medal gains and countries' competitiveness scores. We found that **scores above 20** showed **no significant correlation**, so we **focused on scores below 20** for the great coach impact model. We recommended sports for countries like **CHN**, **GRE**, and **SGP** to invest in coaches, estimating events of most increases in total medals: **wrestling** (2.5), **swimming** (1.58), and **rowing** (0.58).

Our **GSRF-based model** revealed some surprising findings, such as the **higher overall competitiveness of countries** winning medals in **less popular events** compared to more popular ones. Additionally, sports like **table tennis**, **gymnastics**, and **baseball** were historically **dominated by specific regions**.

Finally, we conducted a **sensitivity analysis**, confirming the **robustness** and **reliability** of our models. While the models have strengths and limitations, they provide valuable insights for analyzing **Olympic medal counts**.

**Keywords:** 2028 Los Angeles Olympics, Medal counts, 3D GSRF-based multiple linear regression model, GSRF model, TOPSIS, AHP, Shapley values, rating model, CUSUM method, sensitivity model

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# 1 Introduction

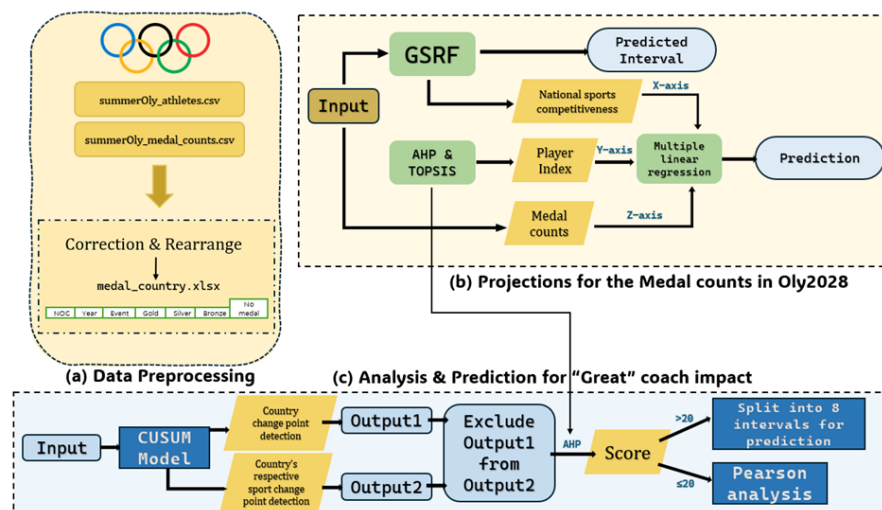
## 1.1 Background

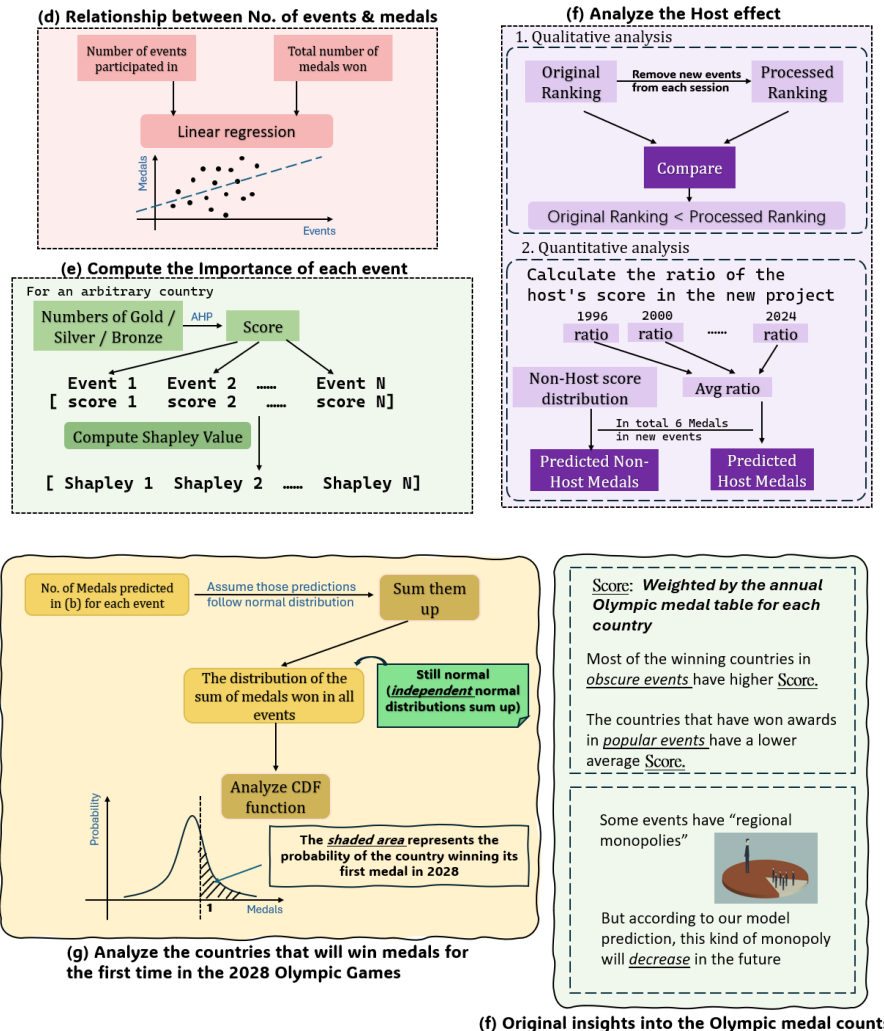
The Olympics, officially known as the Olympic Games, is a major international multi-sport event held every four years, bring friendship to all around the world. The Olympic motto is "Citius, Altius, Fortius," which means "Faster, Higher, Stronger," reflecting the spirit of competition and the pursuit of athletic excellence.

## 1.2 Restatement of the Problem

- **Task 1:** Build a model to make projections for the medal table in Los Angeles during the Los Angeles, USA Summer Olympics in 2028, including prediction intervals for all results. Evaluate the future performance of countries at 2028's Olympics.
- **Task 2:** Estimate the possible number of countries that will earn their first medal in the next Olympics and give the odds for the estimation.
- **Task 3:** Find the relationship between the events and the number of medals countries earn and explain what sports carry the most significant role in various countries. In addition, state how the events chosen by the home country impact the result.
- **Task 4:** Examine the data for evidence of changes that might be caused by a great coach effect and estimate how such an effect contributes to medal counts. Choose three countries and identify sports where they should consider investing in a great coach and evaluate that impact.
- **Task 5:** Describe our new insights about Olympic medals counts based on our model and explain how these insights can inform Olympic committees.

## 1.3 Our Work





(f) Original insights into the Olympic medal counts

## 2 Assumptions and Notations

### 2.1 Model Assumptions

With respect to multiple circumstances for modeling, we make the following assumptions:

- **Assumption 1:** Every country that participated in the Olympics would continue to participate in the Olympics of 2028.

**Justification:** It is required to predict future by referencing historical statistics from the past Olympics.

- **Assumption 2:** In task 1, For a country that has not participated in Olympic sports for four years, we count their prediction value in the GSRF model as zero. In addition, for a country with participation in the Olympics less than three times, we take the mean value of their number of medals as the prediction value.

**Justification:** Enough data is required for value prediction in GSRF model.

- **Assumption 3:** We assume the weighting of gold, silver, bronze, and no medal to be the value we set.

**Justification:** In task 1, the weighting of golden, silver, and bronze medals should be arranged according to their importance to calculate the sports strength of each country. Also, to allow the prediction in task 2, we are required to not just consider the number of medals, but also the number of no medals.

### 3 Data Preprocessing

#### 3.1 Data Cleaning

**Data error:** We fixed some errors in the 2024 Paris Olympics that are illustrated in the provided data set by referencing the official website of Olympic[1].

- Kristen Faulkner, a road cyclist from the United States, won gold and bronze medals in the Women's Road Race and Women's Team Pursuit, respectively. However, the data set provided indicates that she did not receive any medals.
- The data set missed data of Katie Grimes, who is a competitor in the 400m individual medley.
- The data set missed the data of United States' silver medal in the mixed relay of triathlon.

### 4 Model Preperation

#### 4.1 GSRF Model

The Grid-Search Random Forest (GSRF) model is a combination of the random forest and the grid search algorithm.

Grid search is a methodical approach employed for hyperparameter tuning in machine learning models. This technique involves defining a set of hyperparameters along with their respective potential values, subsequently conducting an exhaustive search through all possible combinations of these hyperparameters to identify the optimal configuration that maximizes model performance.

Random Forest model grows and combines multiple decision trees to create a forest. When using Random Forest for classification, each tree gives a prediction by classification or a vote. By aggregating and taking the mean value of the predictions of each tree, the Random Forest model is able to give a general final prediction.

In conclusion, the GSRF model allows us to find the best parameter to gain the accuracy of our model and avoid the overfit or underfit of our prediction.

Here, we think time series could give a rough prediction for the number of medals. So, we use input year as a factor to prediction the number of medals as follows:

$$Year \xrightarrow{GSRF} Sum\ of\ Medals$$

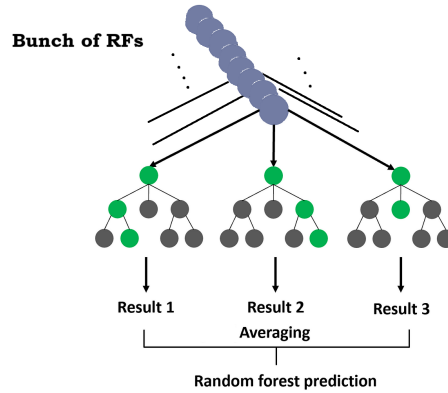


Figure 1: Flow chart of RF

We use this model to do required prediction for arbitrary country and event. That is roughly

$$\text{Number of Events} \times \text{Numbers of Nations} = \text{Number of GSRFs}$$

with regard to the assumption we made on previous section.

However, as the model only includes year as an input and the competitiveness of each country may vary from time to time, we have to develop a more comprehensive prediction model based on the GSRF model to increase its robustness.

## 4.2 Error Analysis on GSRF Model

We use two metrics, MSE and MAE, to find the error in the prediction model. MSE and MAE represent the Mean Squared Error and Mean Absolute Error, respectively. They are defined as follows:

$$MSE := \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2, \quad MAE := \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (1)$$

where  $n$  is the number of samples,  $y$  is the original data and  $\hat{y}$  is the prediction data.

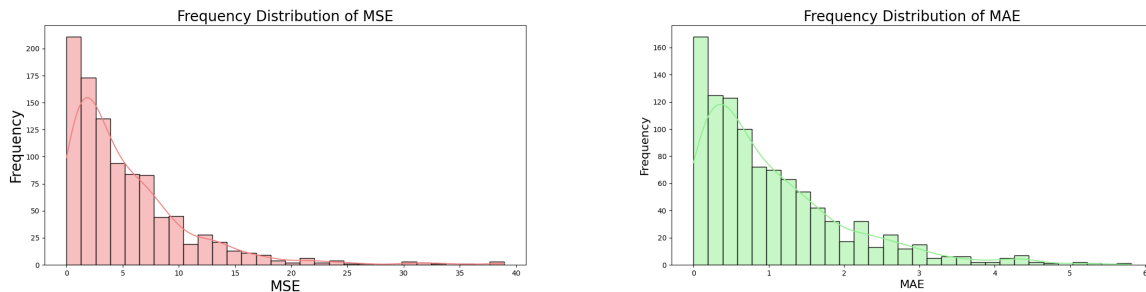


Figure 2: Frequency distribution of MSE and MAE

In figure x, it illustrates that the MSE of most data are concentrated on the interval  $[0, 10]$  and the MAE of most data are concentrated on the interval  $[0, 3]$ . This shows that our GSRF model a

good prediction on the number of medals that each country in each event earned, this allows us to carry on further analysis on developing more comprehensive prediction model.

### 4.3 AHP

The analytic hierarchy process (AHP) is a structured technique for organizing and analyzing complex decisions. It compares the elements at each level of the hierarchy in pairs to determine their relative importance by setting weighting. We use AHP to give a relative significance for golden, silver, bronze, and no medals in order to evaluate the sports strength of a country, it can be summarized by a figure that break down our analysis into three dimensions as follows. We compare

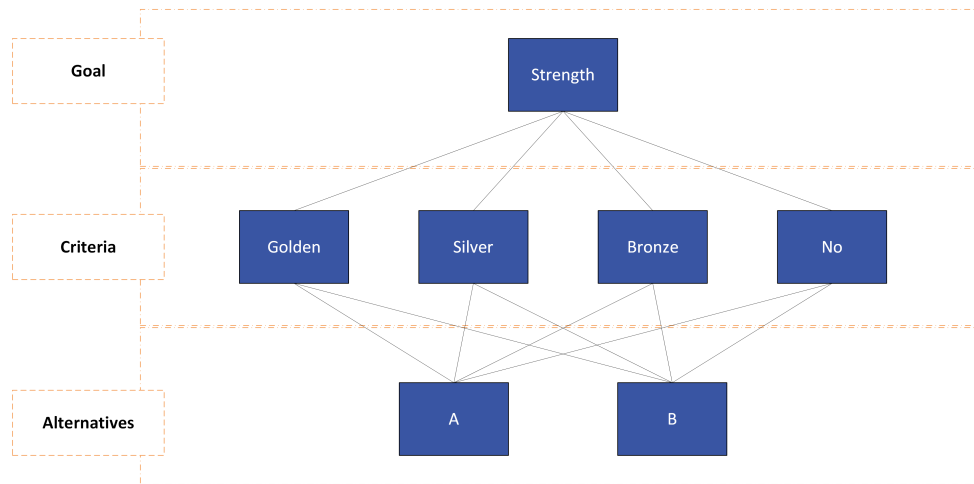


Figure 3: Hierarchical structure model of AHP comparing countries A and B

the four medals and give them a unique weighting one by one for deviating their importance. We set their corresponding weightings on the table below . Note that the required is referencing to a article from The New York Times[2].

|        | Gold  | Silver | Bronze | No |
|--------|-------|--------|--------|----|
| Gold   | 1     | 2      | 4      | 40 |
| Silver | 0.5   | 1      | 2      | 20 |
| Bronze | 0.25  | 0.5    | 1      | 10 |
| No     | 0.025 | 0.05   | 0.1    | 1  |

Table 1: Relative weightings for golden, silver and bronze medals

### 4.4 Consistency Test for AHP

When we determine the weighting, we send two distinct objects into a group, comparing their importance and repeating the process until we get the table of weighting. However, we may ignore the consistency of the whole table. So, we set up a judgment matrix based on the previous table. Then, we use its Consistency Ratio ( $CR$ ) to test whether it is consistent. We first define the

follows:

$$CI := \frac{\lambda_{\max} - n}{n - 1} \quad (2)$$

where  $\lambda_{\max}$  is the maximum eigenvalue of the judgment matrix. Then, we define  $CR$ :

$$CR := \frac{CI}{RI} = \begin{cases} 0, \text{ Judgment matrix is consistent matrix} \\ < 0.1, \text{ Judgment matrix is consistent} \\ \geq 0.1, \text{ Judgment matrix is not consistent matrix} \end{cases} \quad (3)$$

where  $RI$  is random index. Thus, we compute that the consistency index  $CI < 0.10$  which means that the weight determined by AHP is appropriate.

#### 4.5 Data Normalization in AHP

Considering normalization, we make use of geometric mean in order to find the weight vector  $w_i$  with respect to the judgment matrix  $j = (a_{ij})$ . It is described as:

$$w_i = \frac{(\prod_{j=1}^n a_{ij})^{\frac{1}{n}}}{\sum_{k=1}^n (\prod_{j=1}^n a_{kj})^{\frac{1}{n}}} \quad \text{where } i=1,2,\dots,n \quad (4)$$

The required weightings for each medal are shown below:

|            | Golden | Silver | Bronze | No    |
|------------|--------|--------|--------|-------|
| Weightings | 56.338 | 28.169 | 14.085 | 1.408 |

Table 2: Weightings for golden, silver and bronze medals

## 5 Task 1

### 5.1 Problem Analysis for Subtask 1

We only analyze the statistic on 2016, 2020 and 2024 in GSFR model to guarantee its timeliness and did not analyze the new events in 2028 in this section. However, we will still include its statistic on our final predicted medal table as we can use our model in task 3 to make related analysis. By using the model established by GSFR model in the previous section, we could find the predicted number of medals for country  $A$  on event  $B$  in year  $C$  which represent their relative competitiveness. Note that this is **not our final model** because the predicted number of medals should not only based on the medals in the history, but also based on the recent competitiveness of the player in that country, we would add more models based on this model to gain the accuracy. We set it as a  $x$ -axis. Also, by using AHP and Topsis, we can figure out the Topsis score of player  $Y$  (**player's index**) in country  $A$  in event  $B$  in year  $C$  and we set it as a  $y$ -axis. Finally, by setting the actual number of medals that country  $A$  earn in year  $C$  as  $z$ -axis, we can find a regression plane from a three dimensional scatter diagram which gives our final projections for the medal table in the Los Angeles, USA summer Olympics in 2028. Then, we delve into our detailed findings.

## 5.2 Topsis

The Topsis score for  $y$ -value has two versions because we have different weightings when estimating golden medal counts and total medal counts. For the golden medal counts, we derived another set of weighting from AHP to make a more reliable estimation. The exact weightings are 82.47 for gold, 13.74 for silver, 3.43 for bronze, 0.34 for no medal. While for the prediction of total medal counts, its weightings just follow Table 2, and we take it as the example. Also, we create an evaluation matrix  $(p_{ij})_{m \times 4}$  based on the idea of Topsis:

$$\begin{pmatrix} p_{11} & p_{12} & p_{13} & p_{14} \\ p_{21} & p_{22} & p_{23} & p_{24} \\ \vdots & \vdots & \vdots & \vdots \\ p_{m1} & p_{m2} & p_{m3} & p_{m4} \end{pmatrix} = (\mathbf{p}_1, \mathbf{p}_2, \mathbf{p}_3, \mathbf{p}_4) \quad (5)$$

where  $m$  is the number of countries that joined Olympic in both year 2016, 2020 and 2024. We define:

$$\frac{\text{Weighting} \times \text{Number of Golden Medals}}{3} = \text{Golden Index} \quad (6)$$

The vector  $\mathbf{p}_1$  represent the golden index for country  $a_1, a_2, \dots$  in certain event. For example, the first column of  $\mathbf{p}_1$  represent the golden index for country  $a_1$  in event  $C$  in the most three recent Olympic. Then, we consider  $\mathbf{p}_2, \mathbf{p}_3$  and  $\mathbf{p}_4$  represent the silver, bronze, and no medal index for country  $a_2, a_2, \dots$  respectively in in the most three recent Olympic in event  $C$ . Hence, we define:

$$p_i^+ := \max\{p_{1i}, p_{2i}, \dots, p_{mi}\} \quad p_i^- := \min\{p_{1i}, p_{2i}, \dots, p_{mi}\} \quad \text{where } i \in \{1, 2, \dots, n\} \quad (7)$$

$$\mathbf{p}^+ := (p_1^+, p_2^+, p_3^+, p_4^+) \quad \mathbf{p}^- := (p_1^-, p_2^-, p_3^-, p_4^-) \quad (8)$$

where  $\mathbf{p}^+$  is a most ideal point with the highest golden, silver, bronze and no medal index respectively in event  $C$ .  $\mathbf{p}^-$  is a least ideal point with the lowest golden, silver, bronze and no medal index respectively in event  $C$ . Hence, for  $\mathbf{v}_j = (k_1^{(j)}, \dots, k_n^{(j)})$ , we define:

$$d(\mathbf{v}_1, \mathbf{v}_2) := \sqrt{\sum_{i=1}^n (k_i^{(1)} - k_i^{(2)})^2} \quad (9)$$

$$\text{Topsis Score} = S_i := \frac{d(\mathbf{p}_i, \mathbf{p}_i^-)}{d(\mathbf{p}_i, \mathbf{p}_i^+) + d(\mathbf{p}_i, \mathbf{p}_i^-)} \quad (10)$$

Note that  $d(\mathbf{p}_i, \mathbf{p}_i^+)$  is the Euclidean distance between  $p_i$  and the ideal point. Its analogous for  $d(\mathbf{p}_i, \mathbf{p}_i^-)$ . In addition, the calculated weighting for golden medals is 0.82474.

## 5.3 Multivariable Linear Regression Model

Note that we have already constructed our required  $x, y, z$ -axis and so we can build our regression model. Considering a multivariable linear regression model as follows:

$$\begin{cases} \mathbf{Y} = \mathbf{X}\beta + \epsilon \\ E(\epsilon) = 0, \text{cov}(\epsilon, \epsilon) = \sigma^2 I_n \end{cases} \quad (11)$$

We denote the  $k$ - variables linear regression model as  $(\mathbf{Y}, \mathbf{X}\beta, \sigma^2 I_n)$  where

- $\mathbf{Y} = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}$ ,  $\mathbf{X} = \begin{pmatrix} 1 & x_{11} & x_{12} & \cdots & x_{1k} \\ 1 & x_{21} & x_{22} & \cdots & x_{2k} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & x_{n2} & \cdots & x_{nk} \end{pmatrix}$ ,  $\beta = \begin{pmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_k \end{pmatrix}$ ,  $\epsilon = \begin{pmatrix} \epsilon_1 \\ \epsilon_2 \\ \vdots \\ \epsilon_n \end{pmatrix}$
- $y = \beta_0 + \beta_1 \mathbf{x}_1 + \dots + \beta_k \mathbf{x}_k$

Next, we move to solve the required parameter. Utilizing the least squaring method, we make a sum of squares of deviations as:

$$Q = \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_{i1} - \dots - \beta_k x_{ik})^2 \quad (12)$$

As our expected regression plane is  $\hat{\mathbf{z}} = \hat{\beta}_0 + \hat{\beta}_1 \mathbf{x} + \hat{\beta}_2 \mathbf{y}$ , we choose  $\beta_0, \beta_1, \beta_2$  to make  $Q$  the least. Solving the prediction value of  $\beta$ :

$$\hat{\beta} = (\mathbf{X}^T \mathbf{X})^{-1} (\mathbf{X}^T \mathbf{Y}) \quad (13)$$

We get:

$$\hat{\beta}_0 = -1.0120733037064737, \quad \hat{\beta}_1 = 1.024225101, \quad \hat{\beta}_2 = 2.60527834 \quad (14)$$

So, we have our regression plane for the total medals:

$$\hat{\mathbf{z}} = -1.0120733037064737 + 1.024225101\mathbf{x} + 2.60527834\mathbf{y} \quad (15)$$

Similarly, we have our regression plane for the golden medals:

$$\hat{\mathbf{z}}^* = -0.88112518695430307 + 1.06709103\mathbf{x} + 1.55708989\mathbf{y} \quad (16)$$

Utilizing this model, we can give projections for the medal table in the Olympics in 2028, and the other is our predicted golden and total medal table of the top 5 countries. Also, we have regression

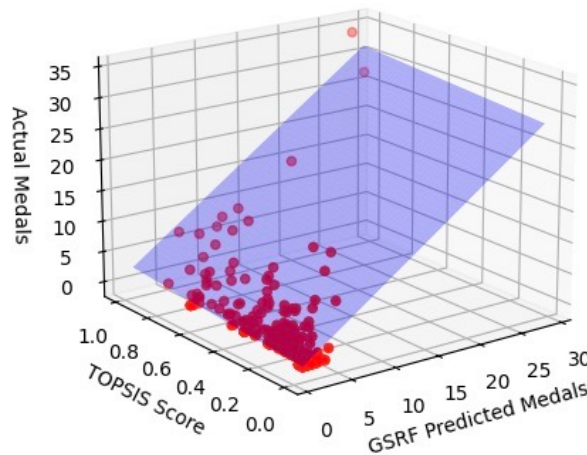


Figure 4: Regression plane

plane in the required three dimensional scatter diagram.

| NOC | Prediction on Total Medals |
|-----|----------------------------|
| USA | 129.5575244                |
| GBR | 99.95260826                |
| CHN | 93.7925119                 |
| FRA | 70.61707962                |
| GER | 68.85432973                |

Table 3: Prediction on total medals

| NOC | Prediction on Golden Medals |
|-----|-----------------------------|
| USA | 45.93389674                 |
| CHN | 36.56557738                 |
| GBR | 33.04866906                 |
| GER | 26.47990924                 |
| AUS | 20.80203945                 |

Table 4: Prediction on golden medals



Figure 5: Left: Word clouds for golden medals Right: Word clouds for total medals

#### 5.4 MSE and Test for Regression Model

For MSE, we refer to equation (1) and we have  $MSE_1 = 0.8696883918278853 < 10$  and  $MSE_2 = 0.9898208651033417 < 10$  which shows that our error in prediction model is small. Then, we introduce the  $R$  test for testing the linear correlation of our regression model. We define:

$$R := \sqrt{\frac{U}{U + Q_e}} \quad (17)$$

where  $U = \sum_{i=1}^n (\hat{y}_i - \bar{y})^2$  and  $Q_e = \sum_{i=1}^n (y_i - \hat{y}_i)^2$ .

We say  $R$  is the multivariable correlation coefficient of  $\hat{\mathbf{z}}$  and  $\mathbf{x}, \mathbf{y}$ . We compute that  $R^2 = 0.9067957932583723$  which is close to 1. Then, we can conclude that  $\hat{\mathbf{z}}$  and  $\mathbf{x}, \mathbf{y}$  has a high linear correlation.

#### 5.5 Prediction Interval

We use GSRF model to find the prediction interval for each country. The GSRF model is based on the model in the model assumptions. However, when the Random Forest model searching for the value predicted by each decision tree, it did not directly yield the final predicted value, it collected each value to form a set  $A$ . We set the quantile of 5% as the lower bound and that of 95% as the upper bound of a subset  $B \subset A$  and the RF model would yield the predicted value of the boundaries. Here is example of the required prediction interval arranged from the value of the upper bound.

| NOC | Predicted Total Medals Lower Bound | Predicted Total Medals Upper Bound |
|-----|------------------------------------|------------------------------------|
| USA | 98.37467893                        | 149.6690602                        |
| CHN | 61.27595238                        | 103.9582143                        |
| GBR | 62.17619048                        | 101.7235265                        |
| GER | 38.33285714                        | 73.15369734                        |
| FRA | 34.59916667                        | 71.97101496                        |
| AUS | 39.72297619                        | 67.52208333                        |
| JPN | 31.28461538                        | 63.04753968                        |

Table 5: Prediction interval of 8 countries

## 5.6 GRA

Based on the predicted medal table on task 1, we use Grey Relation Analysis (GRA) to give the corresponding weighting of golden and total medals. Note that GRA is useful when the dataset size is small (It is small because we only use the data of golden and total medals). Then, we sum up these data to yield the total expected score for each country in 2028's Olympic. Thus, by calculating the difference between the actual score in 2024 and 2028, we can evaluate whether those countries will improve or perform worse.

Next, we move to do the actual calculation based on GRA. We define a referencing sequence  $y = \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}$  where  $y$  represent the properties of a grey system and we take by referring to equation (7):

$$y_1 = \text{Golden Index in 2024}, \quad y_2 = \text{Total Medal Index in 2024} \quad (18)$$

Also, we define a comparative sequence  $x = (x_1 \ x_2)$  and for a country, we take:

$$x_1 = \text{Number of Gold Medals in 2024}, \quad x_2 = \text{Number of Total Medals in 2024} \quad (19)$$

Considering normalization, we define:

$$\tilde{y}_k := \frac{y_k}{\bar{y}_k}, \quad \tilde{x}_k := \frac{x_k}{\bar{x}_k} \quad (20)$$

where  $\bar{y}_k = \frac{y_1 + y_2}{2}$  and  $\bar{x}_k = \frac{x_1 + x_2}{2}$ . Then, for calculating the correlation coefficient between  $x$  and  $y$ , we define:

$$a = \min_i \min_k |x_{1k} - x_{ik}|, \quad b = \max_i \max_k |x_{1k} - x_{ik}| \quad (21)$$

Constructing:

$$\xi_i(k) = \mathbf{y}(x_{1k}, x_{ik}) = \frac{a + 0.5b}{|x_{1k} - x_{ik}| + 0.5b} \quad (22)$$

Then we could evaluate the degree of association in GRA:

$$r_i = \frac{\xi_i(1) + \xi_i(2)}{2} \quad (23)$$

where  $r_1$  is the degree of association (weighting) for  $x_1$  and  $r_2$  is that for  $x_2$ .

We calculate that  $r_1 = 0.5139$  and  $r_2 = 0.4861$ . Therefore, we define the expected score of a country in 2028's Olympic as:

$$\text{Expected Improvement in 2028} = (x_1 r_1 + x_2 r_2) - (\dot{x}_1 r_1 + \dot{x}_2 r_2) \quad (24)$$

where  $\hat{x}_1$  is the number of golden medals of a country in 2024's Olympic and its analogous for  $\hat{x}_2$ . The figure below shows that The United Kingdom (GBR) will has the best improvement and Uzbekistan (UZB) will retrogress the most.

| NOC | Medals | Gold  | Predict | Actual | diff(predict-actual) |
|-----|--------|-------|---------|--------|----------------------|
| UZB | 7.068  | 2.665 | 4.927   | 10.569 | -5.642               |
| KEN | 10.090 | 3.604 | 6.937   | 7.597  | -0.660               |
| BRN | 3.427  | 1.427 | 2.454   | 3.0278 | -0.573               |

Table 6: Least 3 countries with respect to improvement

| NOC | Medals | Gold   | Predict | Actual | diff(predict-actual) |
|-----|--------|--------|---------|--------|----------------------|
| GBR | 99.953 | 33.049 | 67.431  | 38.695 | 28.736               |
| GER | 68.854 | 26.480 | 48.256  | 21.278 | 26.978               |
| BEL | 36.507 | 11.503 | 24.353  | 6.0834 | 18.269               |

Table 7: Top 3 countries with respect to improvement

## 6 Task 2

### 6.1 Motivation and General Idea

In task 1, we have fully constructed a multiple linear regression model to predict medal counts. However, it is worthy of noticing that all predicted value  $\hat{z}$  are on a sports medal counts basis, which makes it quite hard for us to relate total medal counts to concrete probability by using methods of linear regression. Therefore, we utilize the additivity of independent normal distributions and cumulative distribution function (CDF) so as to derive how many countries will get their first medal in 2028 and what is the probability of it. To make sure the idea above is applicable, we need to make following assumptions:

- All the estimated medal counts for each sport are independent.
- Every sports predicted total medal counts follows a normal distribution and assume standard error to be 0.5. Take the value of each estimated total medal counts to be the estimated .
- Mean value of each normal distribution on a sport basis.
- Identify the probability of total medal counts greater than 1 as the probability of whether a country can earn their first medal in history in 2028.

### 6.2 Adding Normal Distribution Up

Due to the additivity of normal distribution, we can get each countrys distribution of total medal counts by adding all their sports medal counts distributions accordingly. Then go through each

country's total medal counts distribution, which are normal distribution with parameters all calculated out, we may get their CDF and find out every country's respective probability of getting the first medal in 2028.

It is obvious that the total number of new medal winners (countries) follows a binomial distribution. And we derive that the expected value is 7.5686. The probability that there will be 7 new winner countries in 2028 is  $p = 0.1537297$ , while for 8 new winner countries,  $p = 0.146856$ .

## 7 Task 3

### 7.1 Linear Fit

Considering the relationship between the events and how many medals countries earn, we first define a threshold of standard score as 3, if some data point exceeds the threshold, we consider it as an outlier and we can filter that outlier to improve the error minimization and improve the crispness of the registered point clouds. Hence, we sketch a linear fit the total number of medals of each country in an arbitrary year and the number of events that the previous country participated in. For example:

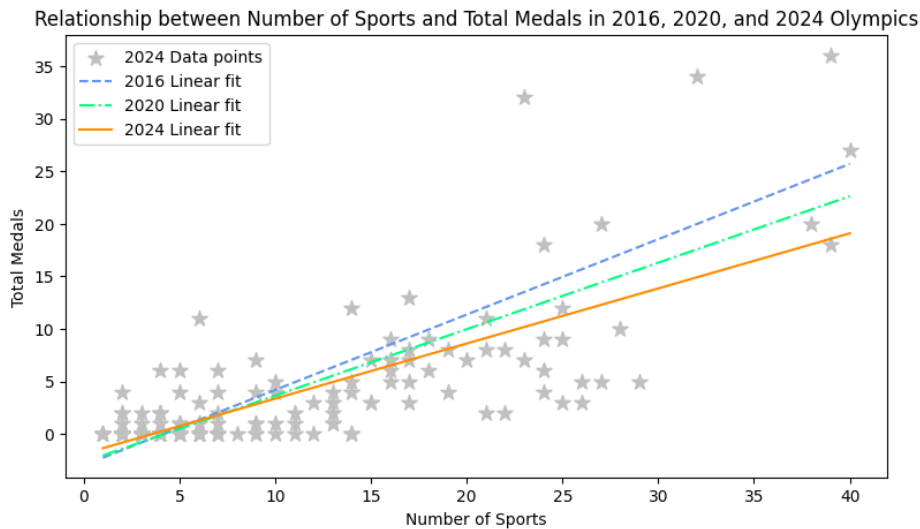


Figure 6: Relationship between number of sports and total medals

### 7.2 Correlation Analysis on Linear Fit

By denoting Pearson and Spearman correlation coefficients as  $r$  and  $r^*$  respectively, we can use  $r$  to find the linear correlation and  $r^*$  to find the monotonic correlation between the number of sports and total medals. The more the  $r$  close to 1, the higher the linear correlation and its the same for  $r^*$ . They are defined as follows:

$$r_{xy} := \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2} \sqrt{\sum_{i=1}^n (y_i - \bar{y})^2}} \quad (25)$$

where

- $n$  is the sample size
- $x_i, y_i$  are the individual sample points indexed with  $i$
- $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$  (the sample mean)

It is analogously for  $\bar{y}$ . Taking 2016, 2020 and 2024 as examples, We have a table below

|                                          | 2016          | 2020          | 2024          |
|------------------------------------------|---------------|---------------|---------------|
| Pearson correlation coefficient $r_{xy}$ | 0.76401666083 | 0.80944543038 | 0.77591128581 |

Table 8: Pearson correlation coefficient in 2016, 2020 and 2024

For the same sample size  $n$ , the  $n$  pairs of raw scores  $(X_i, Y_i)$  are changed into rank variables  $R[X_i], R[Y_i]$  and we have:

$$r_s^* = \rho[[R[X], R[Y]] = \frac{\text{cov}[[R[X], R[Y]]}{\sigma_{R[X]} \sigma_{R[Y]}} \quad (26)$$

where

- $\rho$  denotes the conventional Pearson correlation coefficient operator, but also applied to the rank variables.
- $\text{cov}[R[x], R[Y]]$  is the covariance of the rank variables.
- $\sigma_{R[x]}$  and  $\sigma_{R[Y]}$  are the standard deviations of the rank variables.

|                                          | 2016          | 2020          | 2024          |
|------------------------------------------|---------------|---------------|---------------|
| Spearman correlation coefficient $r_s^*$ | 0.74194731946 | 0.76697999506 | 0.72189174146 |

Table 9: Spearman correlation coefficient in 2016, 2020 and 2024

Combining these two coefficients, it shows that the linear fit has a high correlation to a monotonically increasing linear function.

### 7.3 Shapley Value

We introduce the concept of Shapley value here in order to solve for the relative importance of each sport for various countries. Note that in the Olympic, suppose country  $A$  is good at sport  $X$  with events  $\{x_1, x_2, \dots, x_n\}$ , if events  $\{x_1, x_2, \dots, x_n\}$  all appear in that year's Olympic, country  $X$  tend to invest more capital on developing sport  $X$  and so there will have a joint effect on the number medals that  $A$  earn in  $X$ . This reveals that the "cooperation" between different events is important. Thus, we make use of Shapley value which provides a way to allocate the total gains (or costs) among players based on their individual contributions to the coalition so as to explain

the required relative importance. The Shapley value is described as following: In a colitional game  $(v, N)$  for each member  $i \in N$ ,

$$\text{Shapley Value} = \phi_i(v) = \sum_{S \subset N \setminus \{i\}} \frac{|S|!(n - |S| - 1)!}{n!} (v(S \cup \{i\}) - v(S)) \quad (27)$$

The Shapley value here is calculated based on the weighting in our AHP. For example, for China in 2024's Olympic, we have the related shapley value as follows:

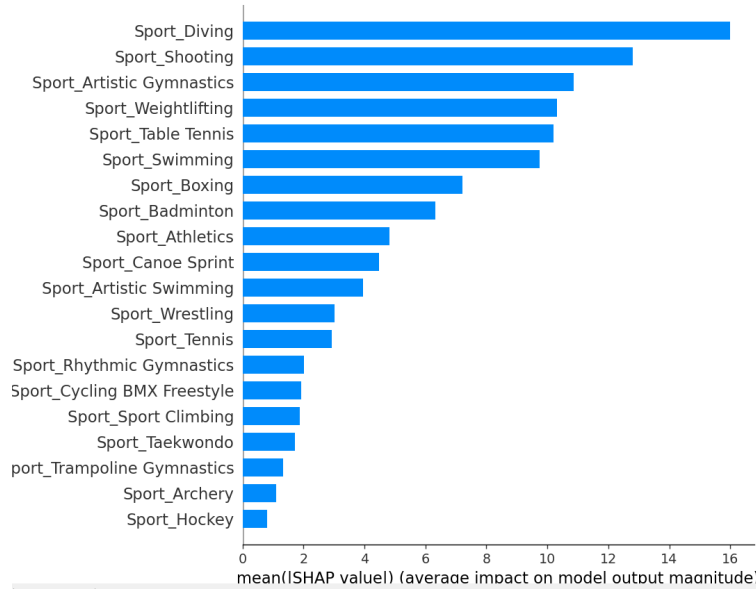


Figure 7: Shapley value of each events in 2024's Olympic with respect to China

This figure shows that diving is the most important sport to China in 2024's Olympic. Using this model, we can also figure out the most important event to other countries. Here is the examples of the most important events for 5 countries:

| NOC | Sport                       | Importance(%) |
|-----|-----------------------------|---------------|
| AFG | Sport_Swimming              | 0.0360125     |
| AIN | Sport_Trampoline Gymnastics | 13.82711068   |
| ALB | Sport_Wrestling             | 1.88496       |
| ALG | Sport_Artistic Gymnastics   | 8.058987823   |
| AND | Sport_Athletics             | 0.10664       |

Table 10: Most important events for 5 countries

## 7.4 Home Country Impact

The home country can add new events in each year's Olympics, this may bring some positive effect on the number of medals that the home country earns. Using our calculated weightings in AHP again, we could compare the host's original rank, the host's rank without new events, and

| Year | Host | Host's Year Total Medals | Host's Year Score | Host's Score Without New Events |
|------|------|--------------------------|-------------------|---------------------------------|
| 2016 | BRA  | 19                       | 944.978           | 939.346                         |
| 2020 | JPN  | 58                       | 2470.329          | 1946.389                        |
| 2024 | FRA  | 64                       | 2263.288          | 2232.303                        |

Table 11: Host's score with and without new events

host's rank without new events according to the weighted score in order to evaluate the home country's impact. Thus, we evaluate the host score in 2016, 2020 and 2024.

Comparing the host rank with and without new events according to the host score, we have the table:

| Year | Host | Host's Year Total Medals | Host's Original Rank | Host's Rank Without New Events |
|------|------|--------------------------|----------------------|--------------------------------|
| 2016 | BRA  | 19                       | 10                   | 28                             |
| 2020 | JPN  | 58                       | 4                    | 11                             |
| 2024 | FRA  | 64                       | 3                    | 4                              |

Table 12: Host's rank with and without new events

Since the host's rank drops if we remove the new events, we can conclude that home country impact brings positive effect on the number of medals for the home country. Below is the heat map that describe the normalized ranking of hosts of eight Olympic from 1996 to 2024 (Note that the diagonal of the map are the host countries):

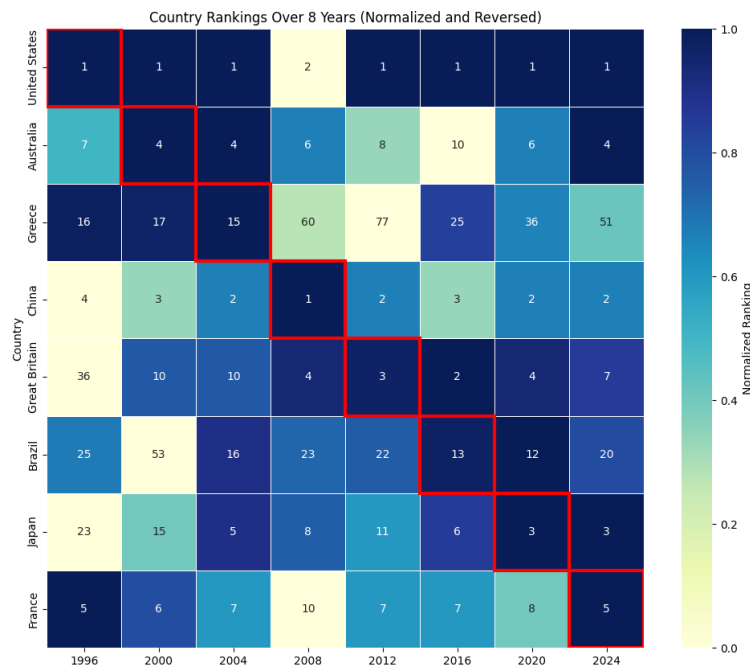


Figure 8: Ranking for each host for eight Olympics

## 8 Task 4

### 8.1 CUSUM

CUSUM algorithm assumes that the mean of the time series between change points are stationary, it calculates the cumulative sum of deviations from a target value. Note that from its assumption, the cumulative sum will also tend to the normal distribution of zero mean, if the cumulative sum deviated from the normal distribution of zero mean from time to time, then it would occur a change point. The algorithm detects the change point by checking whether the cumulative distribution function (CDF) of the sum is smaller than the threshold defined by us. Using the weighting of golden, silver, bronze and no medals in the AHP in task 1, we calculated the weighted score of each country in each event from time to time. By CUSUM algorithm, we get the change points of the weighted score of each country in each event, the figure below shows some statistics of the change points:

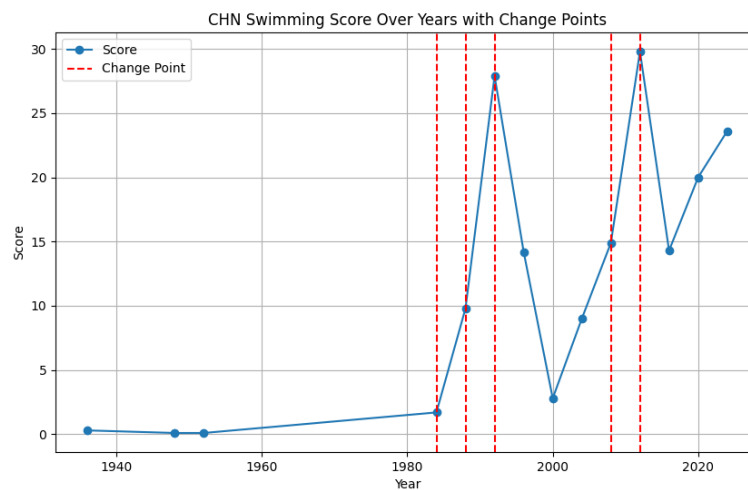


Figure 9: Change Points in CHN swimming score

However, not every change points are due to the impact of Great Coach, for example, we inspect there are some change points in China is due to huge economic growth, home country impact and so on. Since Great Coach Impact should only bring effect on some specific events, for selecting the change point which is resulted by Great Coach Impact, we use CUSUM algorithm to evaluate the change point in the weighted score of each country regardless of specific event.

| ROC | Year | Score |
|-----|------|-------|
| CHN | 2004 | 193.2 |
| CHN | 2008 | 294.4 |

Table 13: CHN's weighted score in 2004 and 2008's Olympic

As the figure indicated above, China has a change point of general weighted score in 2008, combining with the change point of the swimming event in China in 2008, the "change point" may be due to the home country impact instead of the great coach impact. So, we will exclude the

change point of some events with respect to country A in year B, if country A has a corresponding change point in that year B. Here is the example of our filtered statistics of the collected changing point under Great Coach Impact, note that we arrange effect of Great Coach Impact by taking difference between the weighted score which change point located in year  $Y$  and the required weighted score in that Olympic that just before the Olympic in year  $Y$ :

| NOC | Sport      | Year | Score Before | Score After | Score diff |
|-----|------------|------|--------------|-------------|------------|
| USA | Athletics  | 2024 | 63.4         | 91.5        | 28.1       |
| USA | Swimming   | 1960 | 22           | 45.9        | 23.9       |
| USA | Gymnastics | 2004 | 2.3          | 22.2        | 19.9       |
| AUS | Cycling    | 2004 | 12.4         | 32.2        | 19.8       |
| GER | Cycling    | 2000 | 6.2          | 24.6        | 18.4       |

Table 14: Top 5 countries with the greatest effect brought by Great Coach Impact

## 8.2 Interval's Mean

To quantify how the Great Coach Effect contributes to medal counts, we draw a scatter graph, note that the horizontal axis is the score gain of the Olympic that a change point (due to the Great Coach Effect) locate in which we simplified it as "score before". Note that the "score before" for every countries is included here. The vertical axis is the "score diff" which medals gain of the Olympic that a change point locate in:

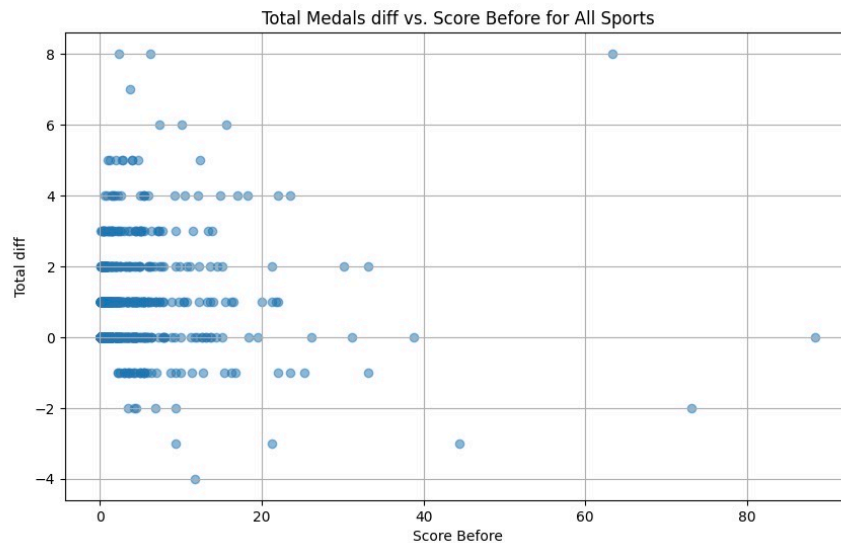


Figure 10: Change points in scatter graph

From this graph, we can see when the score before is greater than 20, the distribution of points is loose. By calculating the Pearson correlation coefficient  $r_{xy}$  for the points with "score before" that greater then 20, we get  $r_{xy} = 0.026887648376916734$  which shows that the great coach did not bring a general effect on those events. Then, for the data point with "score before" less 20, we take eight intervals with equal width (that is  $[0, 2.5]$ ,  $(2.5, 5.0]$ ,  $(5.0, 7.5]$ , ...,  $(17.5, 20]$ ) and we find

the mean of "total diff" for these intervals. If there is no data point on certain interval, we use interpolation method to find the "mean" according to its neighbour intervals. Hence, we get our prediction of Great Coach Effect on each interval, here are some examples (The "total diff" here represent the predicted medals gain):

| Interval     | Total diff  | Sport     |
|--------------|-------------|-----------|
| (2.5, 5.0]   | 1.4         | Athletics |
| (5.0, 7.5]   | 0.777777778 | Athletics |
| (2.5, 5.0]   | 1.666666667 | Fencing   |
| (2.5, 5.0]   | 0.727272727 | Shooting  |
| (5.0, 7.5]   | 1           | Shooting  |
| (15.0, 17.5] | 1           | Fencing   |

Table 15: Total diff of sorted events on the subintervals on  $[0, 20]$

### 8.3 Strategic Coaching Recommendations for Three Countries

To indicate the universality of our model, we would choose three countries with different competitiveness on Olympic which are China (CHN), Greece (GRE) and Singapore (SGP). Note that in 2024's Olympic, China, Greece, Singapore get 91, 8 and 1 medals respectively. Recalled that in the previous section, for "score before" that greater than 20, we consider the "total diff" to be 0. Hence, we can find the events with highest "total diff" so as to make our decision bases on the model developed from previous section. Considering the table, China, Greece, Singapore should

| NOC | Sport      | Score | Total diff  |
|-----|------------|-------|-------------|
| CHN | Wrestling  | 6.7   | 2.5         |
| CHN | Volleyball | 0.1   | 0.833333333 |
| CHN | Water Polo | 0.1   | 0.6875      |
| SGP | Rowing     | 0.1   | 0.586206897 |
| SGP | Sailing    | 1.1   | 0.488636364 |
| SGP | Fencing    | 0.2   | 0.456521739 |
| GRE | Swimming   | 3.4   | 1.583333333 |
| GRE | Athletics  | 5.8   | 0.777777778 |
| GRE | Water Polo | 0.2   | 0.6875      |

Table 16: Top 3 events regarding "total diff" for our chosen countries

invest in wrestling, swimming and rowing in order to enjoy the largest benefit brought by Great Coach Impact.

## 9 Task 5

### 9.1 Relation between the Popularity of Event and Olympic Strength

In this section, we discover that countries that won awards in the least popular event were generally those with stronger Olympic strength. Here, we are interested in which countries win awards

for the most obscure and popular events each year. During data cleaning, we obtained the records the distribution of medals in various countries and events each year. To evaluate different countries, we use the score mentioned earlier to represent a country's overall Olympic strength:

$$Score = 0.56338 \times Gold + 0.28169 \times Silver + 0.14085 \times Bronze + 0.01408 \times No\ medal \quad (28)$$

We calculate the average score[3] (i.e., average comprehensive strength) of all countries from 1968 to 2020. Similarly, through records mentioned previously, we identify the most and least popular events each year and calculate the average scores of countries winning medals in these events. We

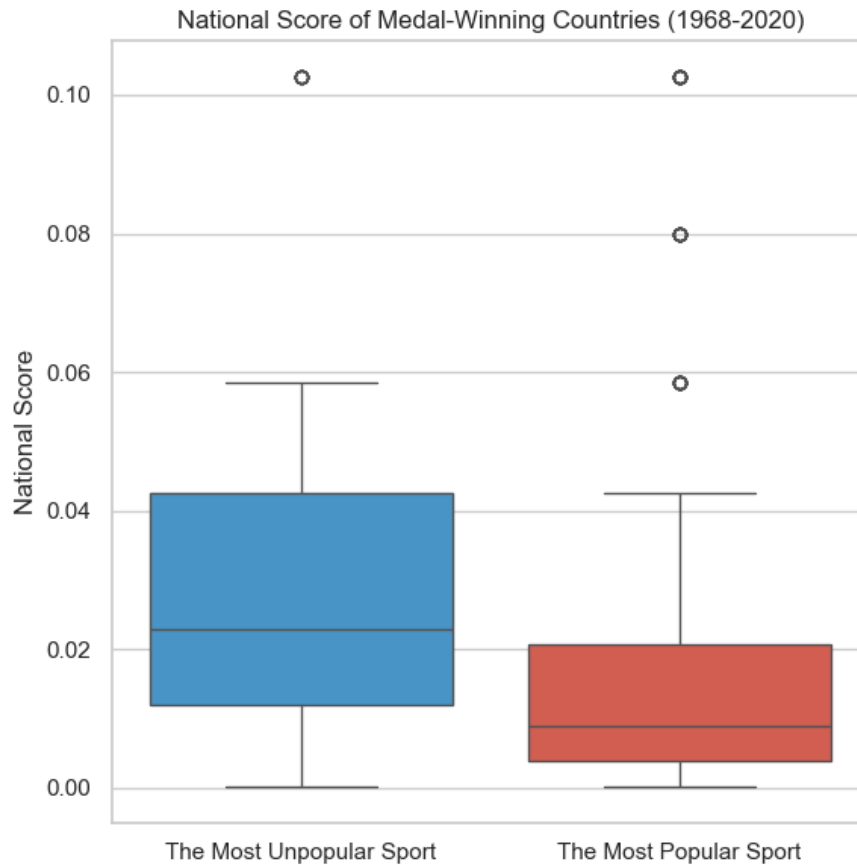


Figure 11: Box and whisker diagram of the comparison of most unpopular sports and most popular sport with respect to average weighted scores

were surprised to find that the countries that won awards in the least popular event were generally those with strong Olympic strength, and vice versa. For the National Olympic Committees of countries with weaker "Olympic strength", I suggest trying to cultivate athletes in less popular events. Due to the low number of participants in less popular events, there is a higher probability of winning the award.

## 9.2 Trend of Monopoly in Olympic Decreasing

Having predicted the medals for various countries and events in 2028 using predictive models, we are curious about future changes in "event monopoly." We used OpenAI to classify all countries into 16 regions, verifying the information through the official website[4]

We then applied the same scoring method as before to calculate the total score for each region in each event from 1980 to 2024, as well as the proportion of scores obtained by each region. We found that three regions had a total score percentage of over 75% for three different events,

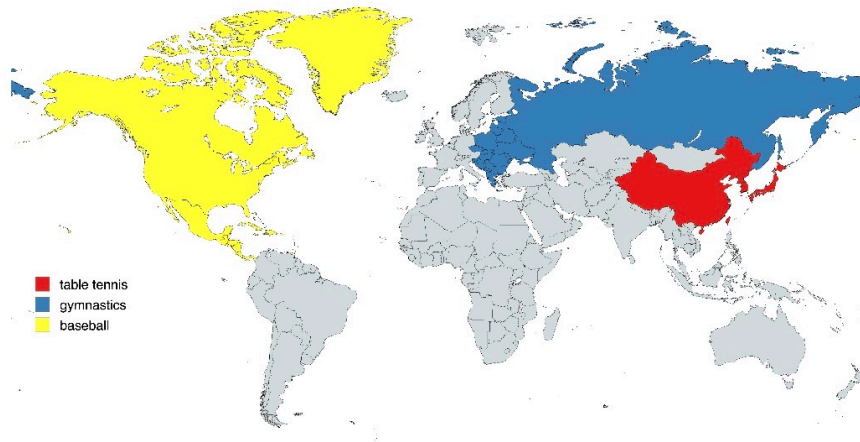


Figure 12: World map that indicate the most monopolized events in East Asia, North America, East Europe that calculated by weighted score

indicating a monopoly. Similarly, we organized the 2028 medal data predicted in the first task. Defining SD as score in dominating regions and SA as score in all regions, we have the figure:

| Region        | Event               | SD | SA | Percentage |
|---------------|---------------------|----|----|------------|
| East Asia     | Table Tennis        | 31 | 35 | 0.8857     |
| North America | Baseball            | 10 | 15 | 0.6667     |
| East Asia     | Badminton           | 22 | 35 | 0.6286     |
| East Europe   | Rhythmic Gymnastics | 5  | 14 | 0.3571     |

Table 17: The proportion of regional scores in relation to the total event score

East Asia still monopolizes table tennis, but North America's monopoly on baseball and East Asia's monopoly on badminton have decreased. The monopoly on artistic gymnastics in Eastern Europe has significantly decreased, likely due to the ban on RUS. Our model predicts a significant decline in Eastern European artistic gymnastics in 2028.

For Olympic Committees in "monopoly areas", we suggest promoting these events to other countries, supporting them, and sharing your experience to maintain a better Olympic competitive environment.

## 10 Sensitivity Analysis

To conduct sensitivity analysis, we introduced Gaussian noise to the GSRFs input. Notice that the only input of GSRF is Year, so we only conducted the sensitivity analysis on Year. Gaussian noise is defined as follows:

$$p(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right) \quad (29)$$

where  $\sigma$  denotes the standard deviation, which is set to be 0.1,  $\mu$  denotes the mean, which is set to be 0. To perform the sensitivity test, we introduced the Gaussian noise according to the probability distribution given by 100 times. The following is the result chart of the sensitivity analysis: We

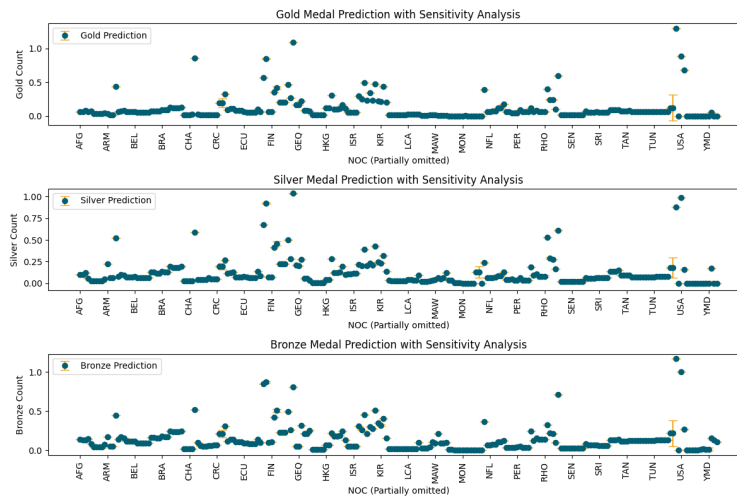


Figure 13: Golden, silver and bronze medal prediction with sensitivity analysis

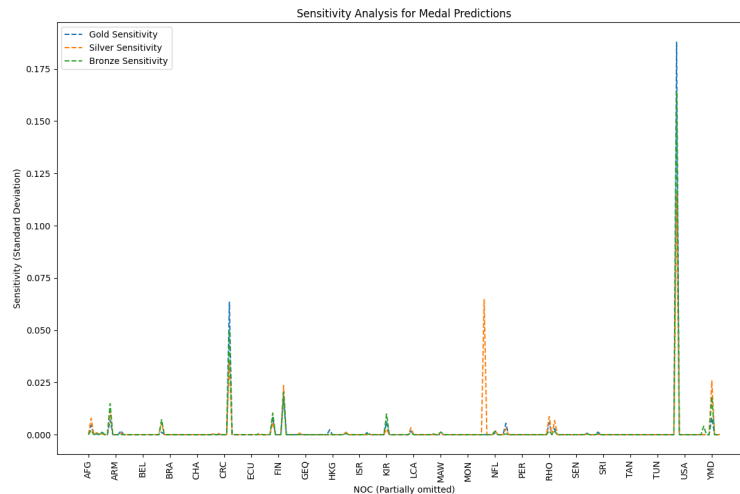


Figure 14: Sensitivity analysis for medal prediction

conducted a sensitivity analysis on three GSRFs for predicting gold, silver, and bronze medals separately. It can be seen from Figure 1 that the error bars of the prediction results of each country

are within a reasonable range. Figure 2 separately extracts the standard deviations of gold, silver, and bronze medal predictions for each country and still within a reasonable range. We speculate that the fluctuation in the results of the country with the highest standard deviation in the graph may be due to the training dataset being too small. Overall, our GSRF model maintains a low sensitivity to changes in Year, indicating strong robustness of the model.

## 11 Model Evaluation

Strengths:

- GSRF: The GS algorithm can automatically tune parameters to ensure optimal performance of the model. And RF averages the results of multiple decision trees, improving the accuracy and stability of the model. We use a GSRF model for each Olympic event, and all prediction results not only rely on the overall data of previous Olympic events, but also learn the changing trends of each event, so as to provide accurate predictions.
- CUSUM: The CUSUM algorithm is highly sensitive to small data changes, making it very effective in monitoring changes in Olympic medal counts over the years. At the same time, the algorithm has strong robustness to noise and outliers in the data.
- TOPSIS and AHP: By constructing a judgment matrix and calculating feature vectors through AHP, we systematically determined the weights of medals, and then used TOPSIS to analyze the weighted number of medals, objectively obtaining the indicators for evaluating athletes

Weakness:

- When analyzing the "great" coach effect, we simply believe that in the absence of a sudden increase in national total scores, the sudden increase in a certain event's performance comes from the "great" coach, but in fact, there may be other factors, such as the participation of talented athletes.
- When predicting the 2028 medal table, we used the GSRF model, which is a machine learning model that requires a large dataset for learning. However, the amount of data given in the question is very small, resulting in poor stability and accuracy of individual results predicted by GSRF

## References

- [1] <https://archive.nytimes.com/beijing2008.blogs.nytimes.com/2008/08/23/the-medal-rankings-which-country-leads-the-olympics/>:
- [2] <https://archive.nytimes.com/beijing2008.blogs.nytimes.com/2008/08/23/the-medal-rankings-which-country-leads-the-olympics/>:
- [3] <https://pandas.pydata.org/>:
- [4] OpenAI ChatGPT (Jan 27, 2025 version, ChatGPT-4):

## 12 AI Report

### 12.1 OpenAI ChatGPT (Jan 27, 2025 version, ChatGPT-4)

Query1: Please help me classify the country codes in the NOC column of Excel by region <Excel File Attached>

It needs to be divided into North America, South America, Northern Europe, Western Europe, Eastern Europe, North Africa, East Africa, West Africa, Central Africa, South Africa, Central Asia, East Asia, South Asia, West Asia, Southeast Asia, and Australia.

Output: Based on the 212 country/region codes you provided, I have classified these countries by region for you. Please note that some countries/regions may span multiple regions or have different historical backgrounds, so I have organized them based on geographical location and the classification principles of the International Olympic Committee. North America: AIA, ANT, CAN, CUB...

### 12.2 OpenAI ChatGPT (Jan 25, 2025 version, ChatGPT-4)

Query1: We want to use a weight based on the number of Olympic gold, silver, and bronze medals to calculate a comprehensive indicator representing a country's Olympic strength. What method should I use to determine this weight.

Output: To calculate a comprehensive indicator representing a country's Olympic strength based on the number of gold, silver, and bronze medals, you'll want to use a method that not only considers the number of medals won but also accounts for the relative importance of each type of medal (gold, silver, bronze). Here is a simple method for determining the weight: Weighted Sum Method (Linear Approach)

This is the simplest and most straightforward approach. You assign a specific weight to each type of medal (gold, silver, bronze) based on their perceived importance. A typical approach is to give higher weight to gold medals, followed by silver, and then bronze.

Example Weights:

- Gold Medal: Weight of 4
- Silver Medal: Weight of 2
- Bronze Medal: Weight of 1

This could reflect the increasing prestige associated with each medal, with gold being the most prestigious. Then, you sum the weighted medals for each country:  $Olympic\ Strength = 4 \times (Gold\ Medals) + 2 \times (Silver\ Medals) + 1 \times (Bronze\ Medals)$   $Olympic\ Strength = 4 \times (Gold\ Medals) + 2 \times (Silver\ Medals) + 1 \times (Bronze\ Medals)$ . This method is simple but still reflects the higher value placed on gold medals.

Why Use This Method:

- Its easy to understand and implement.

- Provides a clear ranking based on a combination of all three medal types.
- Gives countries with more gold medals higher rankings.

### 12.3 OpenAI ChatGPT (Jan 26, 2025 version, ChatGPT-4)

Query1: I want to analyze the contribution of each sub indicator in the overall index to the population. Is there any way to quantify the contribution

Output: The Shapley value is a powerful method from cooperative game theory that helps quantify the contribution of each player (or feature, in your case) to the overall outcome.

In the context of analyzing Olympic strength, the Shapley value can be used to determine how much each sub-indicator (gold, silver, bronze medals, and population size) contributes to the overall Olympic strength index, considering the interaction between these features. "What is the Shapley Value? ... How to Apply Shapley Values to Your Olympic Strength Index ... How to Apply Shapley Values to Your Olympic Strength Index ..."